Gravitational field strength

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Learning target: Use a gravitational field to predict the force on a test mass.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 3.*
Mass 3 kg; weight 15 N toward the source. Weight depends on field.

02. Worked example: explain the model
Field 3 N/kg remains the same; forces differ with test mass and point along the field.

03. Read and interpret the model
Arrows toward source; acceleration follows local inward field.

04. Apply the physics
4, 8, 12 N; field remains 4 N/kg.

05. Explain your reasoning
Force grows with mass but dividing by same mass gives a=g; ideal acceleration independent of test mass.

06. Review: connect an earlier idea *Review: Unit 1, Day 16.*
Net 32 N right; acceleration 8 m/s² right. Leftward motion initially slows.

07. Exit check
g 5 N/kg; 20 N.

08. STRETCH: test the model
Example 1 and 3 kg give 5 and 15 N. Field does not triple when test mass triples.

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01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 3.*
Mass 3 kg; new weight 15 N.

02. Worked example: explain the model
Field 3 N/kg remains the same; forces differ with test mass and point along the field.

03. Read and interpret the model
Arrows toward source; acceleration follows local inward field.

04. Apply the physics
g 3 N/kg; $r=\sqrt{(12/3)}=2$ m.

05. Explain your reasoning
Force grows with mass but dividing by same mass gives a=g; ideal acceleration independent of test mass.

06. Review: connect an earlier idea *Review: Unit 1, Day 16.*
Downslope weight 19.6 N; net 8 N down; a=2 m/s² down. Normal force≈33.95 N.

07. Exit check
2 N/kg; 1 N toward source.

08. STRETCH: test the model
At 2 r; force 0.5 N toward the source.